Mm01a membrane walkthrough

Panama Blot Analyzer 0.2.0 | Western and Protogold | 18 samples

This guide reproduces the analysis saved in **Mm01a-Western-Protogold.panama**. I ran these operations through Panama's analysis engine and reviewed the proposed lanes and landmarks. The steps below give the equivalent on-screen controls.

To inspect the completed experiment: double-click **Open Mm01a experiment.cmd** in the program folder. Use **Save as...** before experimenting with a copy. Follow steps 1-9 to rebuild the analysis from the TIFFs.

1 Import the two membranes

Click **New**, then **01 Images & lanes**. Choose **Western > Import image** and select **Mm01aW.tif**. Choose **Protogold > Import image** and select **Mm01aP.tif**. Both belong in the same gel.

For both images, use **Signal: Dark bands** and **White level: 255**. Keep their original orientation.

2 Place and review the lanes

Set **Lane count: 18** and click **Detect lanes**. Samples run left to right. Inspect the Western overlays, especially faint lanes and the left edge. The fitted strips were about **33.05 pixels** wide, with centers from **46.33 to 1294.67**. Edit coordinates under **Selected lane** if needed.

Protogold strips are paired automatically: **17 strips between the Western lanes**, with the final strip shared by samples 17 and 18, following the Nirvana convention. Inspect them with **Protogold**. Click **Mark lanes reviewed** when satisfied.

3 Crop the scan borders

Click **Auto setup** after reviewing the lanes. This proposes crops, landmarks, baselines and sections. For these images, it kept approximately **2.001668% to 97.998332%** of both profiles. Check the values under **Profile crop**. These exclude the black scan borders; the original TIFFs remain embedded.

4 Replace the automatic protein landmarks

Open **02 Alignment**. Under **Reference > Protein lane**, choose **Sample 10**; also select **Sample 10** in the left sample list. I rejected automatic landmarks near the dirty upper edge and weak incidental extrema.

Using **Remove** (or the row **x**) and **Add fixed**, leave exactly these six landmarks in order: **378, 400, 603, 780, 868, 1125**. Fine-tune each **Position** field. These are positions on the cropped **0-1199 profile axis**, not image pixels or molecular weights.

Align measure and save

5 Match and apply the alignment

Use **Smoothing: 2**, **Prominence: 5%**, **Spacing: 28** and **Search radius: 45**; leave **Enhance protein peaks (FFT)** off. Click **Match lanes**. Select each sample and inspect correspondence with the reference. Correct misplaced matches before clicking **Apply alignment**. The completed trial had **zero unmatched landmarks**.

6 Subtract the background

Open **03 Baseline**. Set **Method: Rolling 10%** and **Window: 121**. Check **Apply to all lanes**, then click **Apply baseline**. Inspect the baseline and corrected traces. This subtracts local background from the aligned Western profiles.

7 Define the shared measurement sections

Open **04 Sections**. Set **Smoothing: 2**, **Prominence: 5%**, **Spacing: 20** and **Max landmarks: 24**, then click **Auto sections**. I removed the first proposal (0-253), changed the next start to **250**, and shortened the final end to **1050**. This excludes the dirty top and lower tail.

Select a section, edit **Name / Start / End**, then **Apply**; use **Remove** to delete one. Rename the remaining sections **Band 01** through **Band 14**. Each consecutive pair below defines one section:

250 | 365 | 388 | 411 | 455 | 475 | 495 | 544
585 | 611 | 676 | 732 | 848 | 884 | 1050

8 Check measurements and run PCA

In **05 Measurements**, choose **Integrated area**: expect **18 samples x 14 sections**. In **06 Analysis**, select **PCA**, keep all 18 samples checked and choose **Measure: Area**, **Normalize: None**, **Scale: Mean center**, with **log(1 + signal)** off. Click **Run PCA**.

Expected variance: **PC1 76.7%** and **PC2 13.3%**. I left biological groups empty and assigned no blanks or intensity standards. These PCA results are exploratory; protein alignment does not normalize sample loading.

9 Save and revisit the experiment

In **Analysis**, click **Excel...** and choose where to save the matrix and results. Click **Save experiment** for a **.panama** file with the TIFFs, edits and completed analyses. Use **Save as...** for a copy and **Load experiment** to reopen it. See the companion **Analysis guide** for heatmaps, LDA, K-means and statistics.

Editing: select a landmark or section and press **Delete**; use **Ctrl+Z** to undo and **Ctrl+Shift+Z** to redo. After landmark edits, match/review and apply alignment again. Rerun PCA after analysis edits, then save.

Check: saving and reopening restored identical measurements and PCA. The settings above are specific to these scans; agreement with historical Nirvana outputs has not yet been established.